

Soil Analysis and Crop Recommendation System

Abstract

This project presents an AI-based system for soil analysis using image processing and machine learning techniques. The system predicts soil nutrient values (Nitrogen, Phosphorus, Potassium) from soil images and recommends suitable crops and fertilizers. The approach combines feature extraction, PCA, and Random Forest models, along with a transformer-based LLM for explanation. The system demonstrates effective prediction and decision support for agriculture.

Introduction

Agriculture productivity depends on soil nutrients. Traditional soil testing is costly and not easily accessible to all farmers. This project aims to provide a low-cost AI-driven solution that uses soil images to estimate nutrient levels and suggest crops and fertilizers.

Literature Review

Previous studies have explored precision agriculture using machine learning and remote sensing. Organizations like ISRO and NASA use satellite data for soil and crop analysis. This project adapts similar concepts using image-based feature extraction and ML models.

Data Exploration

The dataset includes soil image features (RGB and HSV values) and corresponding NPK values. Statistical analysis and preprocessing were performed to understand feature distributions and correlations.

Methodology

The system extracts color features from soil images. These features are scaled using StandardScaler and reduced using PCA. A Random Forest model predicts NPK values. Another ML model predicts suitable crops, and a fertilizer model suggests nutrients. A transformer-based LLM generates human-readable explanations.

Evaluation

The NPK model was evaluated using R2 score, achieving strong predictive performance. Crop and fertilizer models were evaluated using accuracy metrics. The system demonstrated reliable results on test data.

Conclusion

The system successfully predicts soil nutrients and recommends crops and fertilizers. It provides a scalable and cost-effective solution for farmers. Future work includes integrating IoT sensors, improving datasets, and using deep learning for better accuracy.

References

1. Scikit-learn Documentation
2. FastAPI Documentation
3. Research on Precision Agriculture
4. ISRO Agricultural Technologies